HWK 6
Advanced Machine Learning
DATA 642

Objective: This homework assignment focuses on exploring various neural network initialization techniques, activation functions, optimization methods, and regularization strategies. In the first exercise, students will address specific questions related to neural network initialization, biases, activation functions, momentum in optimization, sparse models, and the effects of dropout on training and inference. In the second exercise, students will build a deep neural network (DNN) with five hidden layers, employing He initialization and the ELU activation function, and then train it on a subset of the MNIST dataset (digits 0 to 4) using Adam optimization and early stopping. They will further tune hyperparameters, add batch normalization, and evaluate the impact on learning curves and overfitting, with an additional exploration of dropout's effects. The third exercise involves training a more complex DNN on the CIFAR-10 image dataset with 20 hidden layers, using Nadam optimization and early stopping. Students will compare the impact of batch normalization, SELU activation for self-normalization, and dropout on the model's performance, speed, and convergence, including an evaluation of MC Dropout for achieving better accuracy. Through these exercises, students will gain practical experience in building, tuning, and regularizing deep neural networks for various tasks.

Exercise 1 (3 points)

1. Is it okay to initialize all the weights to the same value as long as that value is selected randomly using He initialization?

2. Is it okay to initialize the bias terms to 0?

3. Name three advantages of the SELU activation function over ReLU.

4. In which cases would you want to use each of the following activation functions: SELU, leaky ReLU (and its variants), ReLU, tanh, logistic, and softmax?

5. What may happen if you set the momentum hyper-parameter too close to 1 (e.g., 0.99999) when using an SGD optimizer?

6. Name three ways you can produce a sparse model.

7. Does dropout slow down training? Does it slow down inference (i.e., making predictions on new instances)? What are about MC dropout?

Exercise 2 (6 points)

Deep Learning.

1. Build a DNN with five hidden layers of 100 neurons each, He initialization, and the ELU activation function.

2. Using Adam optimization and early stopping, try training it on MNIST but only on digits 0 to 4. You will need a softmax output layer with five neurons, and as always make sure to save checkpoints at regular intervals and save the final model so you can reuse it later.

3. Tune the hyper-parameters using cross-validation and see what precision you can achieve.

4. Now try adding Batch Normalization and compare the learning curves: is it converging faster than before? Does it produce a better model?

5. Is the model over-fitting the training set? Try adding dropout to every layer and try again. Does it help?

Exercise 3 (6 points) Practice training a deep neural network (DNN) on the CIRAF10 image dataset.

(a) Build a DNN with 20 hidden layers of 100 neurons each. Use He initialization, and the ELU activation function.

(b) Using Nadam optimization and early stopping, train the network on the CIRAF10 dataset. You can load the dataset through `keras`. The dataset is composed of $60,000$ $32 \times 32-$pixel color images (50,000 for training, 10,000 for testing) with 10 classes, so you will need a softmax output layer with 10 neurons. Remember to search for the right learning rate each time you change the model's architecture or hyper-paramters.

(c) Now, try adding Batch Normalization and compare the learning curves: Is it converging faster than before? Does it produce a better model? How does it affect training speed?

(d) Try replacing Batch Normalization with SELU and make the necessary adjustments to ensure the network self-normalizes (i.e., standardize the input features, use LeCun normal initialization, make sure the DNN contains only a sequence of dense layers, etc.).

(e) Try regularizing the model with dropout. Then without retraining your model, see if you can achieve better accuracy using MC Dropout.